Dhruva Ungrupulithaya

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EDUCATION

NC State University *Raleigh, NC*
PhD and MS in Computer Science *2022 - 2028*

Manipal Institute of Technology *Manipal, India*
Bachelors in Electrical Engineering *2015 - 2019*

SKILLS

Languages: C, C++, Python
Technologies: Operating Systems (Linux), Data Distribution Services (FastDDS), AI/ML, Computer Vision

PUBLICATIONS

[In-Review] Object Aware Compression of LiDAR Point Cloud Data
Ungrupulithaya D, Yoon MK; IEEE/RSJ IROS, Oct. 2026

[In-Review] Kernel-Assisted Integrity Assessment of Camera Streams
Ungrupulithaya D, Yoon MK; ACM Multimedia, Nov. 2026

[In-Review] Evidence Preserving Perception Pipeline for Autonomous Vehicles
Ungrupulithaya D, Zhu Y, Yoon MK; ACM/SIGMOBILE MobiCom, Oct. 2026

RinneFormer: Transformer based Real-World Cooperative Perception Algorithm
Hariharan D, Zhu Y, **Ungrupulithaya D**, Yoon MK, Hollar S; IEEE Intelligent Vehicles, Jun. 2026

Q-Loc: Visual Cue-Based Ground Vehicle Localization Using Long Short-Term Memory
Malinchock C, Yu J, **Ungrupulithaya D**, Thapa P, Yoon MK; IEEE Intelligent Vehicles, Jun. 2025

AdaptAV: Continuous Adaption of Vision Models for Autonomous Vehicles
Zhu Y, **Ungrupulithaya D**, Ge B, Yoon MK; IEEE VTC, Oct. 2024

SELECTED PROJECTS

Adversarial GenAI Attack Detection in Real-Time Camera Streams *Jan 2026 - Apr 2026*

- Designed a robust Linux kernel-assisted application to prevent GenAI based image spoofing, with an accuracy of more than 93% in detecting spoofed images
- Accelerate performance by replacing floating-point arithmetic with fixed-point operations, kernel worker queues, and targeted software optimizations, enabling real-time operation ($<$ 25 ms latency)

Fully Autonomous Delivery Robot *Sep 2025 - Feb 2026*

- Formulated a purely vision based localization technique, reducing capital expenditure by nearly 30%
- Implemented a complete ROS2 based AV stack, with camera and LiDAR perception algorithms, custom path following algorithms, and actuation to enable fully autonomous operation
- Employed large language models (LLMs) to translate live speech into actionable commands, enabling execution of essential locomotion tasks

NCDOT Traffic Analysis *May 2025 - July 2025*

- Analyzed traffic camera data at intersections to identify near misses using YOLO object detection and DeepSORT object tracking, identifying more than 95% of potential accidents

Continuous Model Adaption for Autonomous Vehicles *May 2024 - July 2024*

- Developed a scalable, automated MLOps pipeline utilizing cloud compute to facilitate routine retraining and fine-tuning of vision perception models
- Proposed automated techniques to identify relevant samples, reducing overfitting and improving the model's generalizability, leading to a 20% accuracy improvement

EXPERIENCE

Research and Teaching Assistant — NC State University *Sep 2022 - Present*

- Conducting research on accountable, trustworthy and secure computing for autonomous vehicles
- Created a novel evidence preserving perception pipeline for large scale data logging in autonomous systems, achieving more than 95% reduction in log sizes
- Provide instruction in robotic application development fundamentals, including ROS2, Dockers, computer vision and machine learning for undergraduate and graduate students

Software Developer — KPIT Technologies Ltd. *Feb 2020 - Jul 2022*

- Worked as a technical liaison within a 6-member team building pub/sub communication middleware for a top Japanese automotive OEM, responsible for requirement analysis and technical implementation

- Coordinated with various internal stakeholders to lead the development of a cloud-based annotation tool, resulting in a 60% improvement in productivity of annotation engineers
- Formulated strategies to enhance resource utilization and efficiency of DL applications, resulting in a nearly 50% improvement

Intern — Analog Devices Inc. *Feb 2019 - Aug 2019*

- Developed a test harness to automate the testing and validation of device drivers across 2 families of digital signal processors (DSPs)
- Implemented FreeRTOS kernels in device drivers to support multithreaded real-time applications
- Ported drivers to be compatible across the various DSP families
- Performed thorough evaluation and analysis of various SHARC+ cores to develop next-generation DSPs